EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. If $x\log_e(\log_e x) - x^2 + y^2 = 4 (y > 0)$, then dy/dx at $x = e$ is equal to :

(A) $\frac{e}{\sqrt{4+e^2}}$ (B) $\frac{(1+2e)}{2\sqrt{4+e^2}}$ (C) $\frac{(2e-1)}{2\sqrt{4+e^2}}$ (D) $\frac{(1+2e)}{\sqrt{4+e^2}}$

MDF016

2. If $y = \ell n\left\{\frac{x+\sqrt{(a^2+x^2)}}{a}\right\}$, then the value of $\frac{dy}{dx}$ is-

(A) $\sqrt{a^2-x^2}$ (B) $a\sqrt{a^2+x^2}$ (C) $\frac{1}{\sqrt{a^2+x^2}}$ (D) $x\sqrt{a^2+x^2}$

MDF058

3. Let $u(x)$ and $v(x)$ are differentiable functions such that $\frac{u(x)}{v(x)} = 7$. If $\frac{u'(x)}{v'(x)} = p$ and $\left(\frac{u(x)}{v(x)}\right)' = q$, then $\frac{p+q}{p-q}$ has the value equal to -

(A) 1 (B) 0 (C) 7 (D) –7

MDF059

4. If $y = \sqrt{\frac{1-x}{1+x}}$, then $\frac{dy}{dx}$ equals-

(A) $\frac{y}{1-x^2}$ (B) $\frac{y}{x^2-1}$ (C) $\frac{y}{1+x^2}$ (D) $\frac{y}{y^2-1}$

MDF001

5. If $x = y\,\ell n(xy)$, then $\frac{dx}{dy}$ equals-

(A) $\frac{y(x-y)}{x(x+y)}$ (B) $\frac{x(x+y)}{y(x-y)}$ (C) $\frac{y(x+y)}{x(x-y)}$ (D) $\frac{x(x-y)}{y(x+y)}$

MDF002

6. If $(\cos x)^y = (\sin y)^x$, then $\frac{dy}{dx}$ equals -

(A) $\frac{\log\sin y - y\tan x}{\log\cos x + x\cot y}$ (B) $\frac{\log\sin y + y\tan x}{\log\cos x - x\cot y}$ (C) $\frac{\log\sin y + y\tan x}{\log\cos x + x\cot y}$ (D) $\frac{\log\sin y + y\tan x}{\log\cos y - y\cot x}$

MDF003

7. $\frac{d}{dx}\left(\tan^{-1}\left(\frac{\sqrt{x}-x}{1+x^{3/2}}\right)\right)$ equals- $(x \geq 0)$

(A) $\frac{1}{2\sqrt{x}(1+x)} - \frac{1}{1+x^2}$ (B) $\frac{1}{2\sqrt{x}(1+x)} + \frac{1}{1+x^2}$ (C) $\frac{1}{1+x} - \frac{1}{1+x^2}$ (D) $\frac{1}{1+x} + \frac{1}{1+x^2}$

MDF005

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8. Let f be a function defined for all $x \in R$. If f is differentiable and $f(x^3) = x^5$ for all $x \in R$ $(x \neq 0)$, then the value of $f'(27)$ is-

(A) 15 (B) 45 (C) 0 (D) 35

MDF007

9. If $\Delta_1 = \begin{vmatrix} x & b & b \\ a & x & b \\ a & a & x \end{vmatrix}$ and $\Delta_2 = \begin{vmatrix} x & b \\ a & x \end{vmatrix}$ are given, then -

(A) $\Delta_1 = 3(\Delta_2)^2$ (B) $\frac{d}{dx}\Delta_1 = 3\Delta_2$ (C) $\frac{d}{dx}\Delta_1 = 3(\Delta_2)^2$ (D) $\Delta_1 = 3(\Delta_2)^{3/2}$

MDF008

10. Let $f(x)$ be a polynomial function of second degree. If $f(1) = f(-1)$ and a, b, c are in A.P., then $f'(a), f'(b)$ and $f'(c)$ are in

(A) G.P. (B) H.P. (C) A.G.P. (D) A.P.

MDF009

11. Suppose the function $f(x) - f(2x)$ has the derivative 5 at $x = 1$ and derivative 7 at $x = 2$. The derivative of the function $f(x) - f(4x)$ at $x = 1$, has the value equal to

(A) 19 (B) 9 (C) 17 (D) 14

MDF010

12. If $f(x) = \begin{cases} x\tan^{-1}x + \sec^{-1}(1/x), & x \in (-1,1) - \{0\} \\ \pi/2, & \text{if } x = 0 \end{cases}$, then $f'(0)$ is -

(A) equal to –1 (B) equal to 0 (C) equal to 1 (D) non existent

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MDF011

13. If $x = t^3 + t + 5$ & $y = \sin t$, then $\frac{d^2y}{dx^2} =$

(A) $-\frac{(3t^2+1)\sin t + 6t\cos t}{(3t^2+1)^3}$ (B) $\frac{(3t^2+1)\sin t + 6t\cos t}{(3t^2+1)^2}$

(C) $-\frac{(3t^2+1)\sin t + 6t\cos t}{(3t^2+1)^2}$ (D) $\frac{\cos t}{3t^2+1}$

MDF012

14. Let $f(x) = x + \sin x$. Suppose g denotes the inverse function of f. The value of $g'\left(\frac{\pi}{4} + \frac{1}{\sqrt{2}}\right)$ has the value equal to

(A) $\sqrt{2}-1$ (B) $\frac{\sqrt{2}+1}{\sqrt{2}}$ (C) $2-\sqrt{2}$ (D) $\sqrt{2}+1$

MDF015

15. If $y = x + e^x$ then $\frac{d^2x}{dy^2}$ is :

(A) e^x (B) $-\frac{e^x}{(1+e^x)^3}$ (C) $-\frac{e^x}{(1+e^x)^2}$ (D) $\frac{-1}{(1+e^x)^3}$

MDF013

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16. If $2y=\left(\cot^{-1}\left(\frac{\sqrt{3}\cos x+\sin x}{\cos x-\sqrt{3}\sin x}\right)\right)^2, x\in\left(0,\frac{\pi}{2}\right)$, then $\frac{dy}{dx}$ is equal to :

(A) $2x-\frac{\pi}{3}$ (B) $\frac{\pi}{3}-x$ (C) $\frac{\pi}{6}-x$ (D) $x-\frac{\pi}{6}$

MDF017

17. If $f(1)=1, f'(1)=3$, then the derivative of $f(f(f(x)))+\left(f(x)\right)^2$ at $x=1$ is :

(A) 12 (B) 33 (C) 9 (D) 15

MDF018

18. Suppose that $f(0)=0$ and $f'(0)=2$, and let $g(x)=f\left(-x+f\left(f(x)\right)\right)$. The value of $g'(0)$ is equal to

(A) 0 (B) 1 (C) 6 (D) 8

MDF060

19. Let $f:(-1,1)\to R$ be a differentiable function with $f(0)=-1$ and $f'(0)=1$.
Let $g(x)=[f(2f(x)+2)]^2$. Then $g'(0)$:-

(A) 4 (B) –4 (C) 0 (D) –2

MDF020

20. If $f(4)=g(4)=2$; $f'(4)=9$; $g'(4)=6$ then $\lim\limits_{x\to4}\frac{\sqrt{f(x)}-\sqrt{g(x)}}{\sqrt{x}-2}$ is equal to

(A) $3\sqrt{2}$ (B) $\frac{3}{\sqrt{2}}$ (C) 0 (D) none of these.

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MDF019

21. Let $e^{f(x)}=\ell nx$. If $g(x)$ is the inverse function of $f(x)$, then $g'(x)$ equals to:

(A) e^x (B) e^x+x (C) e^{x+e^x} (D) $e^{x+\ell nx}$

MDF056

22. If $8f(x)+6f\left(\frac{1}{x}\right)=x+5$ and $y=x^2f(x)$, then $\frac{dy}{dx}$ at $x=-1$ is equal to

(A) 0 (B) $\frac{1}{14}$ (C) $-\frac{1}{14}$ (D) None of these

MDF057

23. If $\frac{d^2x}{dy^2}\left(\frac{dy}{dx}\right)^3+\frac{d^2y}{dx^2}=K$, then the value of K is equal to

(A) 1 (B) –1 (C) 2 (D) 0

MDF061

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EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. If $f(x) = (2x - 3\pi)^5 + \frac{4}{3}x + \cos x$ and g is the inverse function of f, then

(A) $g'(2\pi) = \frac{7}{3}$ (B) $g'(2\pi) = \frac{3}{7}$ (C) $g''(2\pi) = \frac{7}{3}$ (D) $g''(2\pi) = 0$

MDF062

2. Let $f(x) = \frac{\sqrt{x - 2\sqrt{x-1}}}{\sqrt{x-1} - 1}.x$, then

(A) $f'(10) = 1$ (B) $f'(3/2) = -1$

(C) domain of $f(x)$ is $x \geq 1$ (D) none

MDF063

3. If $y = x^{(\ell nx)^{\ell n(\ell nx)}}$, then $\frac{dy}{dx}$ is equal to :

(A) $\frac{y}{x}\left(\ell n x^{\ell nx - 1} + 2\ell nx\ \ell n(\ell nx)\right)$ (B) $\frac{y}{x}(\ell nx)^{\ell n(\ell nx)}(2\ell n(\ell nx)+1)$

(C) $\frac{y}{x\ \ell nx}((\ell n\ x)^2 + 2\ \ell n\ (\ell n\ x))$ (D) $\frac{y\ \ell n\ y}{x\ \ell nx}(2\ \ell n\ (\ell n\ x) + 1)$

MDF064

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4. If $2^x + 2^y = 2^{x+y}$, then $\frac{dy}{dx}$ is equal to

(A) $-\frac{2^y}{2^x}$ (B) $\frac{1}{1-2^x}$ (C) $1 - 2^y$ (D) $\frac{2^x(1-2^y)}{2^y(2^x-1)}$

MDF065

5. If $f(x) = \begin{vmatrix} \cos(x+x^2) & \sin(x+x^2) & -\cos(x+x^2) \\ \sin(x-x^2) & \cos(x-x^2) & \sin(x-x^2) \\ \sin 2x & 0 & \sin 2x^2 \end{vmatrix}$, then

(A) $f(-2) = 0$ (B) $f'(-1/2) = 0$ (C) $f'(-1) = -2$ (D) $f''(0) = 4$

MDF066

6. The functions $u = e^x \sin x;\ v = e^x \cos x$ satisfy the equation

(A) $v\frac{du}{dx} - u\frac{dv}{dx} = u^2 + v^2$ (B) $\frac{d^2u}{dx^2} = 2v$

(C) $\frac{d^2v}{dx^2} = -2u$ (D) $\frac{du}{dx} + \frac{dv}{dx} = 2v$

MDF067

7. The differential coefficient of $\sin^{-1}\frac{t}{\sqrt{1+t^2}}$ w.r.t. $\cos^{-1}\frac{1}{\sqrt{1+t^2}}$ is

(A) $1\ \forall\ t > 0$ (B) $-1\ \forall\ t < 0$ (C) $1\ \forall\ t \in R$ (D) none of these

MDF081

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8. If $f(\theta)=\tan\left(\sin^{-1}\sqrt{\frac{2}{3+\cos 2\theta}}\right)$, then-

(A) $f'\left(\frac{\pi}{4}\right)=\sqrt{2}$ (B) $\frac{d(f(\theta))}{d(\cos\theta)}=-2$ at $\theta=\frac{\pi}{4}$

(C) $\frac{d(f(\theta))}{d(\cos\theta)}=-\frac{1}{2}$ at $\theta=\frac{\pi}{4}$ (D) $\frac{d(f(\theta))}{d(\sec\theta)}=1$ at $\theta=\frac{\pi}{3}$

MDF082

COMPREHENSION TYPE QUESTIONS

Paragraph for Question 9 to 11

Let $a_1 > a_2 > a_3 \ldots a_n > 1$; $p_1 > p_2 > p_3 \ldots > p_n > 0$; such that $p_1 + p_2 + p_3 + \cdots \ldots + p_n = 1$. Also $F(x) = \left(p_1 a_1^x + p_2 a_2^x + \ldots\ldots\ldots + p_n a_n^x\right)^{1/x}$ then

9. $\lim\limits_{x\to 0^+} F(x)$ is

(A) $a_1^{p_1} \cdot a_2^{p_2} \ldots a_n^{p_n}$ (B) a_1 (C) a_n (D) does not exist

MDF083

10. $\lim\limits_{x\to \infty} F(x)$

(A) $a_1^{p_1} \cdot a_2^{p_2} \ldots a_n^{p_n}$ (B) a_1 (C) a_n (D) does not exist

MDF084

11. $\lim\limits_{x\to -\infty} F(x)$

(A) $a_1^{p_1} \cdot a_2^{p_2} \ldots a_n^{p_n}$ (B) a_1 (C) a_n (D) does not exist

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MDF085

Paragraph for Question 12 to 14

Consider the function defined implicitly $y^2 + y - x = 2$ on various intervals on the real line. If $y \in \left(-\infty, -\frac{1}{2}\right)$, then equation implicitly defines a unique real valued differentiable function $y = f(x)$. On the basis of above information's, answer the following questions.

12. The value of $f''(4)$ is -

(A) $\frac{2}{125}$ (B) $\frac{125}{2}$ (C) $\frac{1}{5}$ (D) $\frac{1}{125}$

MDF086

13. If $y = g(x)$ is inverse of $y = f(x)$, then $g'(-2)$ is -

(A) $-\frac{1}{3}$ (B) –3 (C) –2 (D) 0

MDF087

14. If $y = h(x)$ is mirror image of $y = f(x)$ about the line $2y + 1 = 0$, then $f''(x_0) + h''(x_0)$, where $x_0 \in \left(-\frac{9}{4}, \infty\right)$ is -

(A) $2x_0$ (B) $\frac{2}{2f(x_0)+1}$ (C) $\frac{2}{(2f(x_0)+1)^3}$ (D) 0

MDF088

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MATCHING LIST TYPE QUESTION

15. Match List-I with List-II and select the correct answer using the code given below the list.

	List-I		**List-II**
(P)	Let $f(x) = x^2 - 4x - 3, x > 2$ and let g be the inverse of f. Find the value of g' at $x = 2$.	(1)	–1
(Q)	Let $f: R \to R$ be defined as $f(x) = x^3 + 3x^2 + 6x - 5 + 4e^{2x}$ and $g(x) = f^{-1}(x)$, then find $g'(-1)$.	(2)	$\frac{1}{6}$
(R)	Suppose f^{-1} is the inverse function of a differentiable function f and let $G(x) = \frac{1}{f^{-1}(x)}$. If $f(3) = 2$ and $f'(3) = \frac{1}{9}$, find $G'(2)$.	(3)	1
(S)	If $y = y(x)$ and it follows the relation $e^{xy} + ycosx = 2$, then $y'(0) + y"(0)$.	(4)	$\frac{1}{14}$

(A) P → 4; Q → 1; R → 2; S → 3 (B) P → 2; Q → 1; R → 3; S → 4

(C) P → 2; Q → 4; R → 1; S → 3 (D) P → 4; Q → 3; R → 2; S → 1

MDF089

16. Match the following List-I with List-II

	List-I		**List-II**
(P)	Let $f(x) = x^2 + xg'(1) + g"(2)$ and $g(x) = x^2 + xf'(2) + f"(3)$, then $f'(1) - f'(2)$ is equal to	(1)	2
(Q)	If $f(x-y), f(x).f(y)$ and $f(x+y)$ are in A.P., for all x, y and $f(0) \neq 0$, then $f'(-2) + f'(2)$ is equal to	(2)	1
(R)	If $f(x) = x^3 + x^2.f'(1) + xf"(2) + f"'(3)$ for all, then $f(0) + f(3) + 1$ is equal to	(3)	4
(S)	Let $f(x) = x^n, n$ being a positive integer, then value of n for which the equality $f'(a+b) = f'(a) + f'(b)$ is valid for all $a, b > 0$ is	(4)	0

(A) P → 4; Q → 3; R → 2; S → 1 (B) P → 2; Q → 4; R → 3; S → 1

(C) P → 3; Q → 4; R → 2; S → 1 (D) P → 1; Q → 2; R → 3; S → 4

MDF090

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EXERCISE (S)

1. At $x=\frac{\pi^2}{4}, \frac{d}{dx}\cos(\sin\sqrt{x}) =$

MDF024

2. If $y\sqrt{x^2+1}=\log\left\{\sqrt{x^2+1}-x\right\}$, then $(x^2+1)\frac{dy}{dx}+xy+1=$

MDF027

3. If $f(x)=x+2$, then $f'(f(x))$ at $x=4$ is

MDF028

4. Let $3f(x)-2f(1/x)=x$, then $4f'(2)$ is equal to

MDF029

5. If $x=2cost-cos2t$ & $y=2sint-sin2t$, find the value of $2\left|\frac{d^2y}{dx^2}\right|$, when $t=(\pi/2)$.

MDF073

6. If $f(x)=mx+c, f(0)=f'(0)=1$ then $f(2)=$

MDF021

7. $\frac{d}{dx}\tan^{-1}(\sec x+\tan x)=\lambda$ then $(10\lambda)=$?

MDF022

8. $\frac{d}{dx}\left[2\tan^{-1}\sqrt{\frac{1-\cos x}{1+\cos x}}\right]=$

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MDF023

9. If $f(x)=\frac{1}{1-x}$, then the derivative of the composite function $f[f\{f(x)\}]$ is equal to

MDF025

10. Let $y=1+\frac{x_1}{x-x_1}+\frac{x_2.x}{(x-x_1)(x-x_2)}+\frac{x_3.x^2}{(x-x_1)(x-x_2)(x-x_3)}+$ ..... upto $(n+1)$ terms

and if $\frac{x}{y}\frac{dy}{dx}=k\left[\frac{x_1}{x_1-x}+\frac{x_2}{x_2-x}+\frac{x_3}{x_3-x}+...+\frac{x_n}{x_n-x}\right]$, then k is

MDF091

11. Let $x=$ cosec θ – sin θ ; $y=$ cosec^6 θ – sin^6 θ if $(x^2+4)\left(\frac{dy}{dx}\right)^2-k(y^2+4)=0$, then k is

MDF092

12. If $x=\frac{1+t}{t^3}$, $y=\frac{3}{2t^2}+\frac{2}{t}$, then $x\left(\frac{dy}{dx}\right)^3-\frac{dy}{dx}$ is equal to

MDF030

13. If $y=ae^{mx}+be^{-mx}$, then $\frac{d^2y}{dx^2}-m^2y=$

MDF068

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14. If $y = \tan^{-1}\frac{1}{x^2+x+1} + \tan^{-1}\frac{1}{x^2+3x+3} + \tan^{-1}\frac{1}{x^2+5x+7} + \tan^{-1}\frac{1}{x^2+7x+13} +$ ..... to 10 terms. Find $(f'(0)+1)^{-1}$.

MDF070

15. If $y = \cot^{-1}\frac{\sqrt{1+\sin x}+\sqrt{1-\sin x}}{\sqrt{1+\sin x}-\sqrt{1-\sin x}}$, find $2\left|\frac{dy}{dx}\right|$ if $x \in \left(0,\frac{\pi}{2}\right)\cup\left(\frac{\pi}{2},\pi\right)$.

MDF071

16. A twice differentiable function $f(x)$ is defined for all real numbers and satisfies the following conditions

$f(0) = 2; f'(0) = -5$ and $f''(0) = 3$.

The function $g(x)$ is defined by $g(x) = e^{ax} + f(x) \quad \forall x \in R$, where '$a$' is any constant.

If $g'(0) + g''(0) = 0$, then number of possible value(s) of 'a' is

MDF072

17. Let $\sqrt{1-x^6}+\sqrt{1-y^6} = a^3.(x^3 - y^3)$ and if $\frac{dy}{dx} = \frac{x^\ell}{y^\ell}\sqrt{\frac{1-y^k}{1-x^k}}$ (where $x \neq \pm 1$), then $\frac{k}{\ell}$ is

MDF093

18. If $f(2) = 4, f'(2) = 1$ then $\lim\limits_{x\to 2}\frac{xf(2)-2f(x)}{x-2} =$

MDF026

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EXERCISE - JEE (Main) PYQ

1. If $x = 3\tan t$ and $y = 3\sec t$, then the value of $\frac{d^2y}{dx^2}$ at $t = \frac{\pi}{4}$, is: **[JEE (Main) 2019]**

(1) $\frac{3}{2\sqrt{2}}$ (2) $\frac{1}{3\sqrt{2}}$ (3) $\frac{1}{6}$ (4) $\frac{1}{6\sqrt{2}}$

MDF035

2. Let $f(x) = \begin{cases} -1, -2 \le x < 0 \\ x^2 - 1, 0 \le x \le 2 \end{cases}$ and $g(x) = |f(x)| + f(|x|)$. Then, in the interval (–2, 2), g is :- **[JEE (Main) 2019]**

(1) differentiable at all points
(2) not differentiable at two points
(3) not continuous
(4) not differentiable at one point

MDF036

3. For $x >$, if $(2x)^{2y} = 4e^{2x-2y}$, then $(1+\log_e 2x)^2 \frac{dy}{dx}$ is equal to : **[JEE (Main) 2019]**

(1) $\log_e 2x$ (2) $\frac{x\log_e 2x + \log_e 2}{x}$ (3) $x\log_e 2x$ (4) $\frac{x\log_e 2x - \log_e 2}{x}$

MDF037

4. If $e^y + xy = e$, the ordered pair $\left(\frac{dy}{dx}, \frac{d^2y}{dx^2}\right)$ at $x = 0$ is equal to : **[JEE (Main) 2019]**

(1) $\left(-\frac{1}{e}, \frac{1}{e^2}\right)$ (2) $\left(\frac{1}{e}, \frac{1}{e^2}\right)$ (3) $\left(\frac{1}{e}, -\frac{1}{e^2}\right)$ (4) $\left(-\frac{1}{e}, -\frac{1}{e^2}\right)$

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MDF038

5. The derivative of $\tan^{-1}\left(\frac{\sin x - \cos x}{\sin x + \cos x}\right)$, with respect to $\frac{x}{2}$, where $\left(x \in \left(0, \frac{\pi}{2}\right)\right)$ is : **[JEE (Main) 2019]**

(1) $\frac{1}{2}$ (2) $\frac{2}{3}$ (3) 1 (4) 2

MDF039

6. Let $y = y(x)$ be a function of x satisfying $y\sqrt{1-x^2} = k - x\sqrt{1-y^2}$ where k is a constant and $y\left(\frac{1}{2}\right) = -\frac{1}{4}$. Then $\frac{dy}{dx}$ at $x = \frac{1}{2}$, is equal to : **[JEE (Main) 2020]**

(1) $\frac{\sqrt{5}}{2}$ (2) $-\frac{\sqrt{5}}{2}$ (3) $\frac{2}{\sqrt{5}}$ (4) $-\frac{\sqrt{5}}{4}$

MDF040

7. If $y(\alpha) = \sqrt{2\left(\frac{\tan\alpha + \cot\alpha}{1+\tan^2\alpha}\right) + \frac{1}{\sin^2\alpha}}, \alpha \in \left(\frac{3\pi}{4}, \pi\right)$, then $\frac{dy}{d\alpha}$ at $\alpha = \frac{5\pi}{6}$ is : **[JEE (Main) 2020]**

(1) 4 (2) $-\frac{1}{4}$ (3) $\frac{4}{3}$ (4) -4

MDF041

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8. If $y=\sum_{k=1}^{6} k\cos^{-1}\left\{\frac{3}{5}\cos kx-\frac{4}{5}\sin kx\right\}$, then $\frac{dy}{dx}$ at $x=0$ is _____. **[JEE (Main) 2020]**

MDF042

9. Let $x^k+y^k=a^k,(a,K>0)$ and $\frac{dy}{dx}+\left(\frac{y}{x}\right)^{\frac{1}{3}}=0$, then k is : **[JEE (Main) 2020]**

(1) $\frac{3}{2}$ (2) $\frac{1}{3}$ (3) $\frac{2}{3}$ (4) $\frac{4}{3}$

MDF094

10. If $\left(a+\sqrt{2}\,b\cos x\right)\left(a-\sqrt{2}\,b\cos y\right)=a^2-b^2$, where $a>b>0$, then $\frac{dx}{dy}$ at $\left(\frac{\pi}{4},\frac{\pi}{4}\right)$ is : **[JEE (Main) 2020]**

(1) $\frac{a-b}{a+b}$ (2) $\frac{a+b}{a-b}$ (3) $\frac{2a+b}{2a-b}$ (4) $\frac{a-2b}{a+2b}$

MDF095

11. Consider the function $f(x)=\begin{cases}\frac{P(x)}{\sin(x-2)} & ; \quad x\neq 2\\ 7 & ; \quad x=2\end{cases}$

Where $P(x)$ is a polynomial such that $P''(x)$ is always a constant and $P(3)=9$. If $f(x)$ is continuous at $x=2$, then $P(5)$ is equal to _____. **[JEE (Main) 2021]**

MDF043

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12. Let $f(x)=\cos\left(2\tan^{-1}\sin\left(\cot^{-1}\sqrt{\frac{1-x}{x}}\right)\right), 0<x<1$. Then : **[JEE (Main) 2021]**

(1) $(1-x)^2 f'(x)-2\left(f(x)\right)^2=0$ (2) $(1+x)^2 f'(x)+2\left(f(x)\right)^2=0$

(3) $(1-x)^2 f'(x)+2\left(f(x)\right)^2=0$ (4) $(1+x)^2 f'(x)-2\left(f(x)\right)^2=0$

MDF044

13. If $y(x)=\cot^{-1}\left(\frac{\sqrt{1+\sin x}+\sqrt{1-\sin x}}{\sqrt{1+\sin x}-\sqrt{1-\sin x}}\right), x\in\left(\frac{\pi}{2},\pi\right)$, then $\frac{dy}{dx}$ at $x=\frac{5\pi}{6}$ is : **[JEE (Main) 2021]**

(1) $-\frac{1}{2}$ (2) -1 (3) $\frac{1}{2}$ (4) 0

MDF045

14. Let $f:S\to S$ where $S=(0,\infty)$ be a twice differentiable function such that $f(x+1)=xf(x)$. If $g:S\to R$ be defined as $g(x)=\log_e f(x)$, then the value of $|g''(5)-g''(1)|$ is equal to : **[JEE (Main) 2021]**

(1) $\frac{205}{144}$ (2) $\frac{197}{144}$ (3) $\frac{187}{144}$ (4) 1

MDF096

15. If $y=y(x)$ is an implicit function of x such that $log_e(x+y)=4xy$, then $\frac{d^2y}{dx^2}$ at $x=0$ is equal to. **[JEE (Main) 2021]**

MDF097

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16. The value of $\log_e 2 \frac{d}{dx}(\log_{\cos x} \operatorname{cosec} x)$ at $x=\frac{\pi}{4}$ is **[JEE (Main) 2022]**

(1) $-2\sqrt{2}$ (2) $2\sqrt{2}$ (3) -4 (4) 4

MDF046

17. Let $x(t)=2\sqrt{2}\cos t\sqrt{\sin 2t}$ and $y(t)=2\sqrt{2}\sin t\sqrt{\sin 2t}$, $t\in\left(0,\frac{\pi}{2}\right)$. Then $\frac{1+\left(\frac{dy}{dx}\right)^2}{\frac{d^2y}{dx^2}}$ at $t=\frac{\pi}{4}$ is equal to **[JEE (Main) 2022]**

(1) $\frac{-2\sqrt{2}}{3}$ (2) $\frac{2}{3}$ (3) $\frac{1}{3}$ (4) $\frac{-2}{3}$

MDF047

18. If $y=\tan^{-1}(\sec x^3-\tan x^3)$. $\frac{\pi}{2}<x^3<\frac{3\pi}{2}$, then **[JEE (Main) 2022]**

(1) $xy''+2y'=0$ (2) $x^2y''-6y+\frac{3\pi}{2}=0$

(3) $x^2y''-6y+3\pi=0$ (4) $xy''-4y'=0$

MDF048

19. Let $f(x)=\begin{vmatrix} a & -1 & 0 \\ ax & a & -1 \\ ax^2 & ax & a \end{vmatrix}$, $a\in R$. Then the sum of the squares of all the values of a for which $2f'(10)-f'(5)+100=0$ is : **[JEE (Main) 2022]**

(1) 117 (2) 106 (3) 125 (4) 136

MDF049

20. If $y(x)=(x^x)^x, x>0$ then $\frac{d^2x}{dy^2}+20$ at $x=1$ is equal to: **[JEE (Main) 2022]**

MDF050

21. If $y(x)=x^x$, $x>0$, then $y''(2)-2y'(2)$ is equal to : **[JEE (Main) 2023]**

(1) $8\log_e 2-2$ (2) $4\log_e 2+2$ (3) $4(\log_e 2)^2-2$ (4) $4(\log_e 2)^2+2$

MDF051

22. Let $y=f(x)=\sin^3\left(\frac{\pi}{3}\left(\cos\left(\frac{\pi}{3\sqrt{2}}\left(-4x^3+5x^2+1\right)^{\frac{3}{2}}\right)\right)\right)$. Then, at $x=1$, **[JEE (Main) 2023]**

(1) $2y'+\sqrt{3}\pi^2y=0$ (2) $2y'+3\pi^2y=0$ (3) $\sqrt{2}y'-3\pi^2y=0$ (4) $y'+3\pi^2y=0$

MDF052

23. Let $y(x)=(1+x)(1+x^2)(1+x^4)(1+x^8)(1+x^{16})$. Then $y'-y''$ at $x=-1$ is equal to **[JEE (Main) 2023]**

(1) 976 (2) 464 (3) 496 (4) 944

MDF053

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24. Let $f(x)=\frac{\sin x+\cos x-\sqrt{2}}{\sin x-\cos x}, x\in[0,\pi]-\left\{\frac{\pi}{4}\right\}$. Then $f\left(\frac{7\pi}{12}\right)f''\left(\frac{7\pi}{12}\right)$ is equal to **[JEE (Main) 2023]**

(1) $\frac{-2}{3}$ (2) $\frac{2}{9}$ (3) $-\frac{1}{3\sqrt{3}}$ (4) $\frac{-2}{3\sqrt{3}}$

MDF054

25. If $2x^y + 3y^x = 20$, then $\frac{dy}{dx}$ at (2, 2) is equal to **[JEE (Main) 2023]**

(1) $-\left(\frac{3+\log_e 8}{2+\log_e 4}\right)$ (2) $-\left(\frac{2+\log_e 8}{3+\log_e 4}\right)$ (3) $-\left(\frac{3+\log_e 16}{4+\log_e 8}\right)$ (4) $-\left(\frac{3+\log_e 4}{2+\log_e 8}\right)$

MDF055

26. If $y(\theta)=\frac{2\cos\theta+\cos 2\theta}{\cos 3\theta+4\cos 2\theta+5\cos\theta+2}$, then at $\theta=\frac{\pi}{2}$, $y'' + y' + y$ is equal to: **[JEE (Main) 2024]**

(1) $\frac{3}{2}$ (2) 1 (3) $\frac{1}{2}$ (4) 2

MDF148

27. Let $f(x)= x^5 + 2x^3 + 3x + 1, x \in R$, and $g(x)$ be a function such that $g(f(x)) = x$ for all $x \in R$. Then $\frac{g(7)}{g'(7)}$ is equal to : **[JEE (Main) 2024]**

(1) 7 (2) 42 (3) 1 (4) 14

MDF149

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28. If $y=\frac{\left(\sqrt{x}+1\right)\left(x^2-\sqrt{x}\right)}{x\sqrt{x}+x+\sqrt{x}}+\frac{1}{15}\left(3\cos^2 x-5\right)\cos^3 x$, then $96y'\left(\frac{\pi}{6}\right)$ is equal to : **[JEE (Main) 2024]**

MDF150

29. Let for a differentiable function $f:(0,\infty)\to R$, $f(x)-f(y)\geq\log_e\left(\frac{x}{y}\right)+x-y$, $\forall\, x, y \in (0,\infty)$. Then $\sum_{n=1}^{20} f'\left(\frac{1}{n^2}\right)$ is equal to ____. **[JEE (Main) 2024]**

MDF151

30. Suppose

$$f(x)=\frac{\left(2^x+2^{-x}\right)\tan x\sqrt{\tan^{-1}\left(x^2-x+1\right)}}{\left(7x^2+3x+1\right)^3},$$

Then the value of $f'(0)$ is equal to **[JEE (Main) 2024]**

(1) π (2) 0 (3) $\sqrt{\pi}$ (4) $\frac{\pi}{2}$

MDF152

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EXERCISE - JEE (Advanced) PYQ

1. Let $g(x) = \ell n f(x)$ where $f(x)$ is a twice differentiable positive function on $(0, \infty)$ such that $f(x+1) = xf(x)$. Then for $N = 1, 2, 3, \ldots\ldots$, $g''\left(N + \frac{1}{2}\right) - g''\left(\frac{1}{2}\right) =$ **[JEE (Advanced) 2008]**

(A) $-4\left\{1 + \frac{1}{9} + \frac{1}{25} + \ldots\ldots + \frac{1}{(2N-1)^2}\right\}$ (B) $4\left\{1 + \frac{1}{9} + \frac{1}{25} + \ldots\ldots + \frac{1}{(2N-1)^2}\right\}$

(C) $-4\left\{1 + \frac{1}{9} + \frac{1}{25} + \ldots\ldots + \frac{1}{(2N+1)^2}\right\}$ (D) $4\left\{1 + \frac{1}{9} + \frac{1}{25} + \ldots\ldots + \frac{1}{(2N+1)^2}\right\}$

MDF074

2. Let f and g be real valued functions defined on interval $(-1, 1)$ such that $g''(x)$ is continuous, $g(0) \neq 0, g'(0) = 0, g''(0) \neq 0$, and $f(x) = g(x)\sin x$.

Statement-1 : $\lim\limits_{x \to 0} [g(x)\cot x - g(0)\text{cosec}x] = f''(0)$

and

Statement-2 : $f'(0) = g(0)$

[JEE (Advanced) 2008]

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation of statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

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MDF075

3. If the function $f(x) = x^3 + e^{\frac{x}{2}}$ and $g(x) = f^{-1}(x)$, then the value of $g'(1)$ is

[JEE (Advanced) 2009]

MDF076

4. Let $f(\theta) = \sin\left(\tan^{-1}\left(\frac{\sin\theta}{\sqrt{\cos 2\theta}}\right)\right)$, where $-\frac{\pi}{4} < \theta < \frac{\pi}{4}$. Then the value of $\frac{d}{d(\tan\theta)}(f(\theta))$ is

[JEE (Advanced) 2011]

MDF077

5. Let $f : \mathbb{R} \to \mathbb{R}$, $g : \mathbb{R} \to \mathbb{R}$ and $h : \mathbb{R} \to \mathbb{R}$ be differentiable functions such that $f(x) = x^3 + 3x + 2$, $g(f(x)) = x$ and $h(g(g(x))) = x$ for all $x \in \mathbb{R}$. Then- **[JEE (Advanced) 2016]**

(A) $g'(2) = \frac{1}{15}$ (B) $h'(1) = 666$ (C) $h(0) = 16$ (D) $h(g(3)) = 36$

MDF079

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	C	A	B	B	B	A	A	B	D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	A	A	C	B	D	B	C	B	A
Que.	21	22	23							
Ans.	C	C	D							

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B,D	A,B	B,D	A,B,C,D	B,C,D	A,B,C,D	A,B	A,D	A	B
Que.	11	12	13	14	15	16				
Ans.	C	A	B	D	C	C				

EXERCISE (S)

1.	0	**2.**	0	**3.**	1	**4.**	2	**5.**	3.00
6.	3	**7.**	5	**8.**	1	**9.**	1	**10.**	1.00
11.	36.00	**12.**	1	**13.**	0	**14.**	101	**15.**	1.00
16.	2	**17.**	3.00	**18.**	2				

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	4	4	1	4	2	1	91	3	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	39	3	1	1	40	4	4	2	3	16
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	3	2	3	2	2	4	4	105	2890	3

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5
Ans.	A	A	2	1	B,C

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Important Notes

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